

Probability Distributions



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- ① Probability Distribution: The possible values a variable can take and how frequently they occur.

↳ Y = Actual Outcome of an Event
 y = One of the possible outcome.

$$P(Y=y) \text{ or } P(y)$$

e.g. Y = No. of Red marbles drawn out of a bag.

$$y = 5.$$

$$\therefore P(Y=5) \text{ or } P(5)$$

↳ Since $P(y)$ = Probability of each distinct outcome

$$\therefore P(y) = \text{Probability function}$$

↳ Probability Distributions measure the likelihood of an outcome depending on how often it is featured in the Sample Space.



① Probability Distributions are defined using two main characteristics:

- 1) Mean : Average Value (μ)
- 2) Variance : How Spread out data is.
(σ^2) (How far away from mean all the values are)
- 3) Standard Deviation (σ) : Square Root of Variance.

② Relationship between Mean & Variance:

A constant relationship between mean & Variance exists for any distribution.

Variance equals the Expected Value of the Squared difference from the mean for any value.

$$\sigma^2 = E((Y - \mu)^2)$$

$$\sigma^2 = E(Y)^2 - \mu^2$$

① Types of Probability Distributions:

① Discrete Distributions:

↳ Uniform Distribution

↳ Bernoulli Distribution

↳ Binomial Distribution

↳ Poisson Distribution

② Continuous Distributions:

↳ Normal Distribution

↳ Student's - T Distribution

↳ Chi-Squared Distribution

↳ Exponential Distribution

↳ Logistic Distribution.



① Discrete Distribution :

Shows all possible values of a discrete random variable along with the associated probabilities.

Have finitely many distinct outcomes.

$$P(Y \leq y) = P[Y < (y+1)]$$

$$\text{i.e. } P(Y \leq 3) = P(Y < 4)$$

(1) Uniform Distribution :

Denoted by 'U'.

$U(a, b)$; a, b : range of values in data set

$X \sim U(3, 7) \Rightarrow$ Variable X follows

Discrete Uniform Distribution
ranging from 3 to 7.

Events which follow Uniform Distribution are ones where all outcomes have equal probability.

e.g. Rolling a single, standard, six sided die.

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

One drawback of Uniform Distribution is that Expected Value provide no relevant information.

i.e. All outcomes have same probability.

<2> Bernoulli Distribution :

Denoted by ~~Bernoulli~~ 'Bern'.

$X \sim \text{Bern}(p)$; Variable X Follows a Bernoulli Distribution with a probability of Success = p .



Any Event where we have 1 Trial and 2 possible Outcomes Follows Bernoulli Distribution.

e.g. Coin flip

$$\text{Variance } \sigma^2 = p(1-p)$$

$$\text{Standard Deviation } \sigma = \sqrt{p(1-p)}$$

↳ Binomial Distribution:

A Sequence of Identical Bernoulli Events.

Denoted by 'B'.

$$X \sim B(n, p) ; \begin{aligned} X &= \text{Variable} \\ n &= \text{no. of trials} \\ p &= \text{probability of Success.} \end{aligned}$$

$\therefore$ Variable X Follows a Binomial Distribution with n trials and a likelihood of ' p ' to succeed on each individual trials.



e.g. $X \sim B(10, 0.6)$

$$\therefore \text{Bern}(p) = B(1, p)$$

The Binomial Distribution is the Discrete Probability Distribution that gives only two possible results in an Experiment, either Success or Failure.

In Binomial Probability Distribution, the number of Successes in a Sequence of 'n' Experiments, where each time a question is asked in yes/no, then the boolean valued outcome is represented either with Success / yes / true / one (probability p) or

Failure / ~~False~~ No / False / zero (probability $q = 1 - p$)

e.g. Quiz Consisting of 10 true or False questions.



So, guessing a single true or False question is a Bernoulli Event.

Guessing the entire quiz is a Binomial Event.

E (Bernoulli Event) $\rightarrow$ Which outcome we expect for a single trial.

E (Binomial Event) $\rightarrow$ Number of times we expect to get a specific outcome.

P (desired Outcome) $= p$

P (Alternative Outcome) $= q = 1 - p$

$$p(y) = {}^nC_y \cdot p^y \cdot (1-p)^{n-y}$$

Binomial

Distribution



e.g. A single Stock of general motors:

$$\begin{aligned} p(\text{up}) &= 0.6 \\ p(\text{down}) &= 0.4 \end{aligned}$$

$\therefore$ Z Increases in 5 days:

$$P(3) = {}^5C_3 \cdot (0.6)^3 \cdot (0.4)^{5-3}$$

$$= \frac{5!}{3!(5-3)!} \cdot 0.216 \cdot 0.16$$

$$= \cancel{10} (10) (0.216) (0.16)$$

$$= 0.3456$$

$$\therefore = 34.56\%$$

Expected Value:

$$E(x) = x_0 \cdot P(x_0) + x_1 \cdot P(x_1) + \dots + x_n \cdot P(x_n)$$

For Variable $Y \sim B(n, p)$



$$E(Y) = p \cdot n$$

$$\text{Variance } \sigma^2 = E(Y^2) - E(Y)^2$$

$$= n \cdot p \cdot (1-p)$$

ex: If a coin is tossed 5 times,
Find the probability of
a) Exactly 2 Heads,
b) At least 4 Heads,
3) At least 2 Heads.

↳ No. of trials $n = 5$

Probability of Head = Probability of Success
 $= \frac{1}{2}$

Probability of Tail = Probability of Failure
 $= \frac{1}{2}$

$$\therefore p = \frac{1}{2}, q = \frac{1}{2}$$



a) Exactly 2 Heads :

$$X = 2$$

$$\therefore P(X=2, 0.5) = \frac{5!}{2!(5-2)!} \cdot (0.5)^2 \cdot (0.5)^{5-2}$$

$$= 0.3125$$

b) Atleast 4 Heads :

$$X \geq 4$$

$$\therefore P(X \geq 4) = P(X=4) + P(X=5)$$

$$P(X=4) = \frac{5!}{4!(5-4)!} \left(\frac{1}{2}\right)^4 \cdot \left(\frac{1}{2}\right)^{5-4}$$

$$= \frac{5}{32}$$

$$P(X=5) = \frac{5!}{5!(5-5)!} \cdot \left(\frac{1}{2}\right)^5 \cdot \left(\frac{1}{2}\right)^{5-5}$$

$$= \frac{1}{32}$$



$$\therefore P(X \geq 4) = \frac{5}{32} + \frac{1}{32}$$

$$= \frac{6}{32} = \frac{3}{16}$$

c) At least 2 Heads :

$$X \geq 2$$

$$P(X \geq 2) = P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= 1 - [P(X=0) + P(X=1)]$$

$$= 1 - \left[\frac{5!}{0!(5-0)!} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{5-0} \right] +$$

$$\left[\frac{5!}{1!(5-1)!} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{5-1} \right]$$

$$= 1 - \left[\frac{1}{32} + \frac{5}{32} \right] = \frac{13}{16}$$

4) Poisson Distribution:

Denoted by 'Po'.

$Y \sim \text{Po}(\lambda)$; Variable Y Follows
a poisson
Distribution with
 λ frequency.

$$P(Y) = \frac{\lambda^Y \cdot e^{-\lambda}}{Y!} ; e = \text{Euler's number}$$

OR
Napier's Constant
 $e \approx 2.72$

Expected value $E(Y) = \lambda$

Variance = λ

Poisson Distribution is a Discrete Probability Function that means the variable can only take Specific Values in a given Range, probable Infinite.



Poisson Distribution measures how many times an Event is likely to occur within 'x' period of time.

A poisson random variable 'x' defines the number of Success in the Experiment.

The Distribution occurs when there are Events that do not occur as the outcomes of a definite number of Outcomes.

Poisson Distribution is used under Certain Conditions:

- ↳ Number of trials 'n' tends to infinity.
- ↳ Probability of Success 'p' tends to zero.

In the Binomial Distribution, the Probability Distribution of the number of times an outcome occurred in a fixed number of trials was considered.



Thus, the Variable had an upper limit in Binomial Distribution.

However, an Upper Limit on the number of opportunities for an Outcome to Occur Cannot be logically Fixed.

e.g. No. of Black spots on ~~a~~ a piece of Cloth.

No. of Potential Visitors to a Departmental Store.

Poisson Distribution can be used in Such Situations.

⇒ Requisites to Use Poisson Distribution:

(1) The numerous opportunities for the occurrence of the outcome even when an upper limit exists, is a very large value ($n \rightarrow \infty$), with a very low probability of occurrence of the outcome during an opportunity. ($p \rightarrow 0$)



If ' X ' denotes the Random variable indicating the number of occurrences which follows a Poisson Distribution, the probabilities of ' X ' taking a given value

$P(X) = 0, 1, 2, \dots$ depends on its expected value λ .

λ is the parameter of the distribution.

When X follows a Poisson Distribution with Parameter λ :

$E(X) = \lambda$; Expected Value of X is λ .

$Var(X) = \lambda$; Variance of X is λ .

Std. deviation $(X) = \sqrt{\lambda}$

Ex: The number of Spares required for any component in a machine during the machine's life follows a Poisson Distribution



With $\lambda = 2$.

The firm that uses this machine, must decide how many spares of this component are needed so that the probability of a stock out is at most 0.06.

↳ Let the variable X denote the number of spares required.

X follows a Poisson Distribution with parameter $\lambda = 2$.

Poisson Distribution table values for $\lambda = 2$ and probability of $X \geq x$ i.e. Cumulative probabilities for each value of X .

x	$\lambda = 2$	Look for Poisson Distribution table online for these values.)
0	0.1353	
1	0.4060	
2	0.6767	
3	0.8571	
4	0.9473	
5	0.9834	
6	0.9955	



Here, $P(X \leq x) = \sum_{r=0}^x \frac{\lambda^r \cdot e^{-\lambda}}{r!}$

Now, Stock out probability for a given value of x is

x	0	1	2	3	4	5
$P(X \leq x)$	0.1353	0.4060	0.6767	0.857	0.9473	0.9834
Stock out Probability	0.8647	0.594	0.3233	0.1429	0.0527	0.0166

↓
6

0.9955

0.0046

For Stock-Out probability to be at most 0.06, $x = 4$.

$\therefore$ Firm should have 4 spares of the component.

Ex: A random variable ' x ' has a poisson distribution with parameter λ , such that,



$$P(X=1) = 0.2 \cdot P(X=2)$$

Find $P(X=0)$.

↳ For Poisson Distribution:

$$P(X=x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$P(X=1) = \frac{e^{-\lambda} \cdot \lambda^1}{1!}$$

$$\therefore 0.2 \cdot P(X=2) = \lambda \cdot e^{-\lambda}$$

$$\therefore 0.2 \cdot \frac{e^{-\lambda} \cdot \lambda^2}{2!} = \lambda \cdot e^{-\lambda}$$

$$\therefore \frac{\lambda^2}{10} = \lambda$$

$$\therefore \lambda = 10$$

$$P(X) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$\therefore P(X=0) = \frac{e^{-10} \cdot 10^0}{0!} = e^{-10}$$



Ex: Telephone calls arrive at an exchange according to poisson Distribution at a rate $\lambda = 2/\text{min}$.

Calculate the probability that exactly two calls will be received during each of the first 5 mins of the hour.

$\rightarrow$ Let 'N' = No. of calls received during 7 min.

$$P(N=2) = \frac{e^{-\lambda} \cdot \lambda^2}{2!}$$

$$\therefore P(N=2) = 2 \cdot e^{-2}$$

Now, "M" be the number of minutes among 5 minutes considered, during which exactly 2 calls will be received.

$\therefore$ M Follows a Binomial Distribution with Parameter $n=5$ and $p = 2e^{-2}$.



$$P(M=5) = 32 \cdot e^{-10}$$

$$\therefore P(M=5) = 0.00145$$

② Continuous Distributions

A Continuous Variable takes any value in a Certain Range.

e.g. Height, weight, Volume, Duration.

↳ Sample Space is infinite.

↳ We cannot record the frequency of each distinct value.

↳ Graph of frequency distribution is called 'Probability Density Function'.

↳ It is generally accepted that probability of any individual value from a continuous distribution is essentially zero.



$$\therefore P(X) = 0$$

$$P(X > X) = P(X \geq X)$$

i.e. A student running a mile in less than 6 min.

$$P(X < 6) = P(X \leq 6)$$

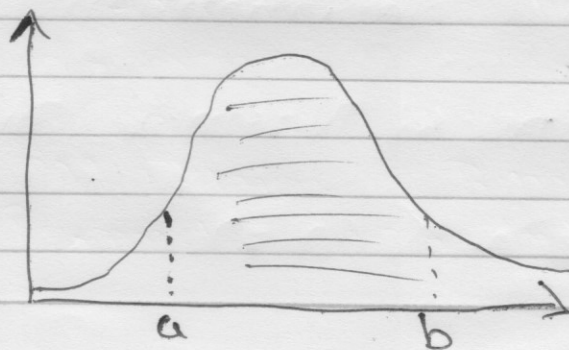
because ~~$P(X = 6)$~~ $P(X = 6) = 0$.

↳ Cumulative Density Function (CDF):

It gives probability ~~upto a~~ by encompassing everything upto a certain value.

$$F(y) = P(Y \leq y)$$

The CDF is especially useful when we want to estimate the probability of some interval.

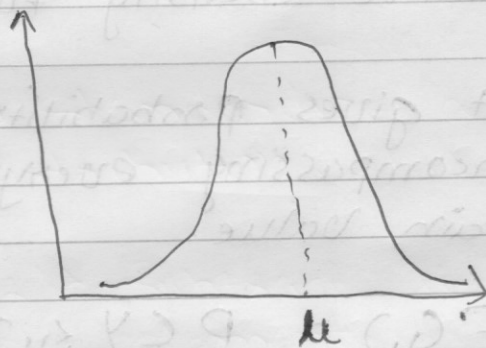


$$P(b > X > a)$$

↳ Normal Distribution :

$X \sim N(\mu, \sigma^2)$; Variable X
Follows a
Normal Distribution
with mean μ and variance σ^2 .

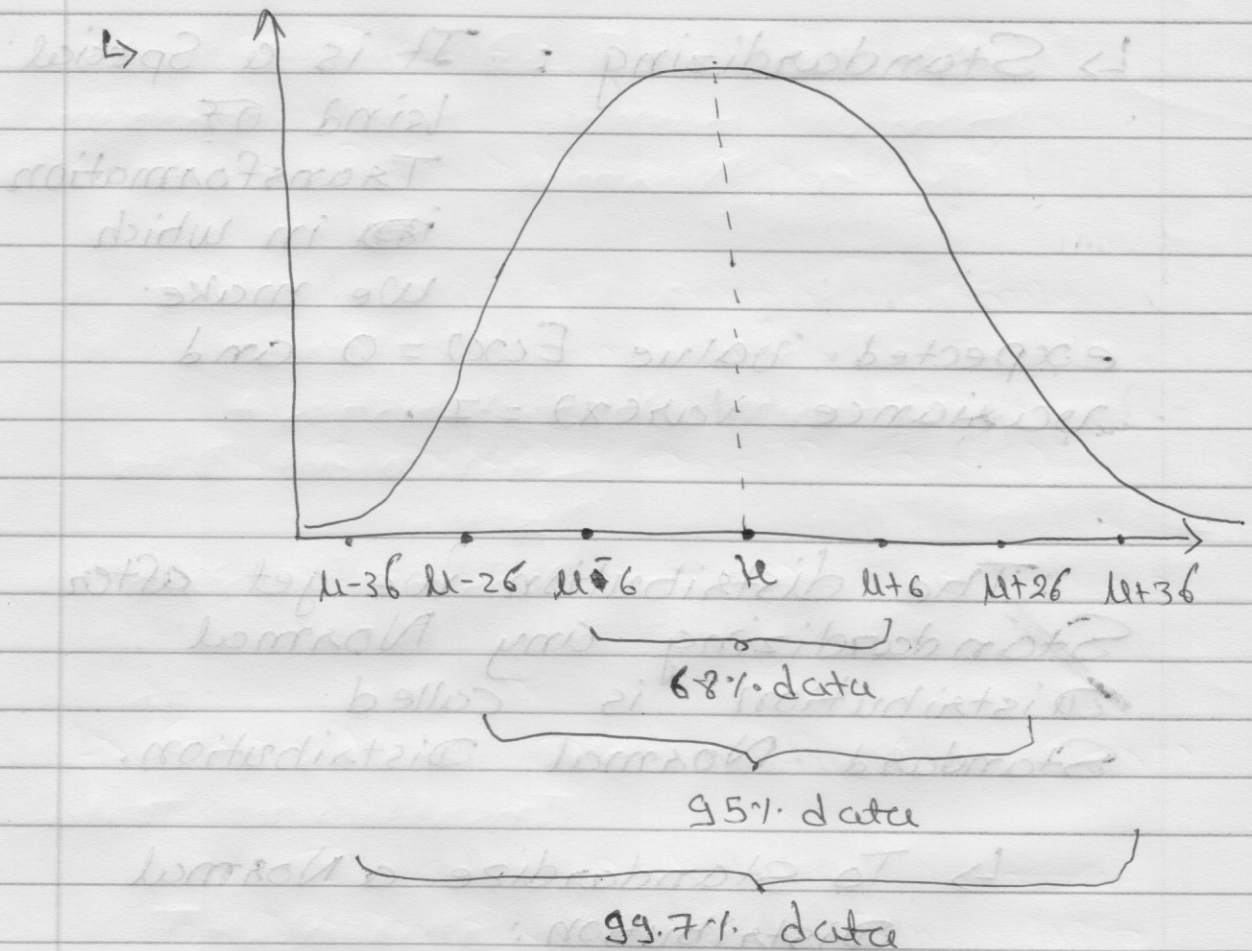
↳ Graph of Normal Distribution is
Bell Shaped and Symmetric.



↳ Expected Value $E(X) = \mu$

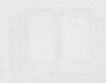
↳ Variance Value $= \sigma^2$





(2) Standard Normal Distribution:

↳ Transformation: A way in which we can alter every Element of a Distribution to get a new Distribution with similar characteristics.



↳ Standardizing : It is a Special kind of Transformation in which we make expected value $E(x) = 0$ and variance $Var(x) = 1$.

The distribution we get after Standardizing any Normal Distribution is called Standard Normal Distribution.

↳ To Standardize a Normal Distribution :

1) Subtract μ from every element.

2) Divide every element of the newly obtained Distribution by the value of the Standard deviation.

↳ Standard Normal Distribution :

$$Z \sim N(0, 1)$$



↳ Importance of Standard Normal Distribution:

Every element of the non standardized distribution is represented in the new distribution by the number of standard deviations it is away from the mean.

i.e. if in Normal distribution

$$y = u + 2.3 \cdot (s)$$



In Standard Normal Distribution

$$y = 2.3$$

↳ Student's-T Distribution:

denoted by "t"

$Y \sim t(k)$; Variable Y

Follows a

Student's-T

Distribution

with k degrees of Freedom.



↳ Student's - T Distribution is a Small Sample Size Approximation of a Normal Distribution.

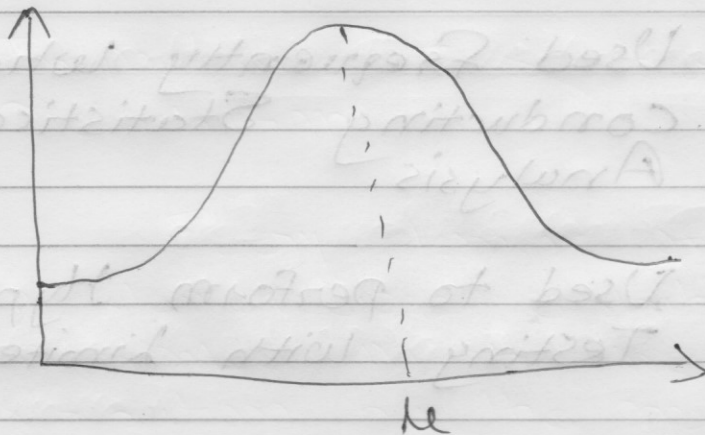
↳ Certain Characteristics + Sufficient Data
= Normal Distribution

Certain Characteristics + Insufficient Data
=

Student's - T Distribution

↳ Graph is Bell shaped and Symmetric

↳ It has fatter tails than Normal Distribution.



↳ Another Difference between Normal and Student's-T Distribution is apart from mean μ and Variance σ^2 , we must also define degrees of Freedom k .

↳ If $k \geq 2$

then

Expected Value $E(y) = \mu$

$$\text{Variance } \text{var}(y) = \frac{s^2 \cdot k}{k-2}$$

s^2 = Variance
of
Sample.

↳ Used frequently when conducting Statistical Analysis.

↳ Used to perform Hypothesis Testing with limited data.

↳ Chi-Squared Distribution :

Denoted by ' χ^2 '; (Capital Greek letter Chi).

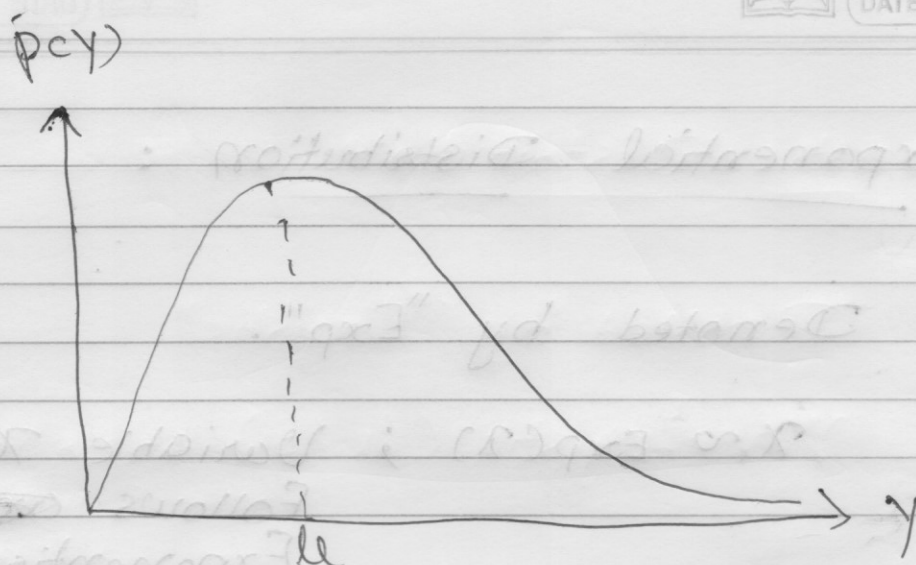
$Y \sim \chi^2(k) \Rightarrow$ Variable Y

Follows a Chi-Squared

Distribution with ' k ' degrees of freedom.

↳ Very few events follow such a distribution.

↳ Used most frequently in Hypothesis testing and Confidence Interval calculations.



↳ Graph is not symmetric.

↳ Highly skewed to right.

↳ Elevating Student's - T Distribution to the Second Power gives us Chi-Squared Distribution and Vice Versa.

$$\begin{array}{l|l} Y \sim t(k) & X \sim \chi^2(k) \\ Y^2 \sim \chi^2(k) & \sqrt{X} \sim t(k) \end{array}$$

↳ Expected Value $E(X) = k$

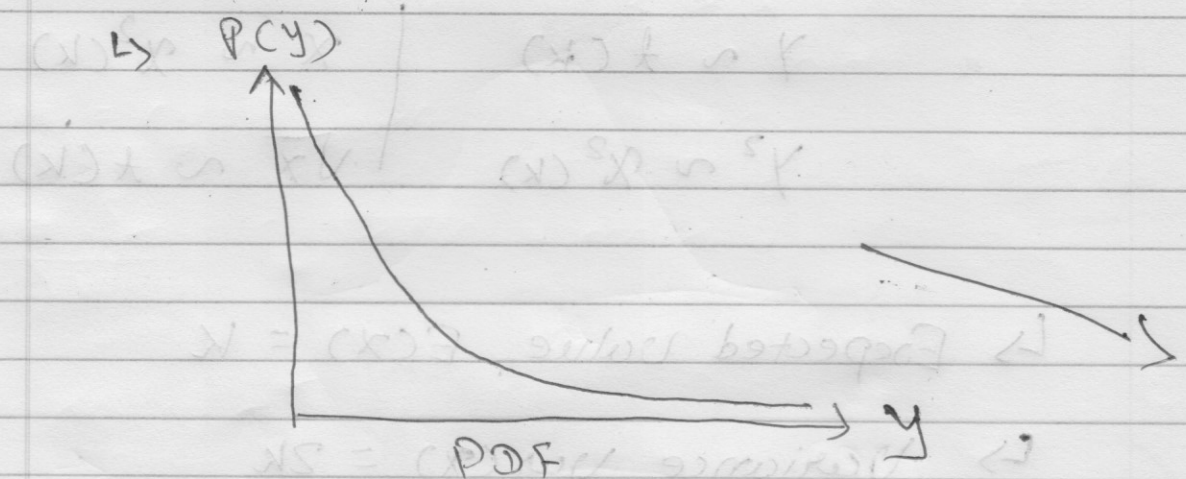
↳ Variance $\text{Var}(X) = 2k$

 Exponential Distribution :

Denoted by "Exp".

$X \sim \text{Exp}(\lambda)$; Variable X
Follows ~~and~~ an
Exponential
Distribution with a scale of λ .

↳ Probability starts off high,
initially decreases,
Eventually plateauing.



↳ ~~Q~~ Graph resembles a Boomerang.



↳ Rate Parameter λ determines how fast the CDF Curve reaches the point of ~~the~~ Plateauing and how spread out the graph is.

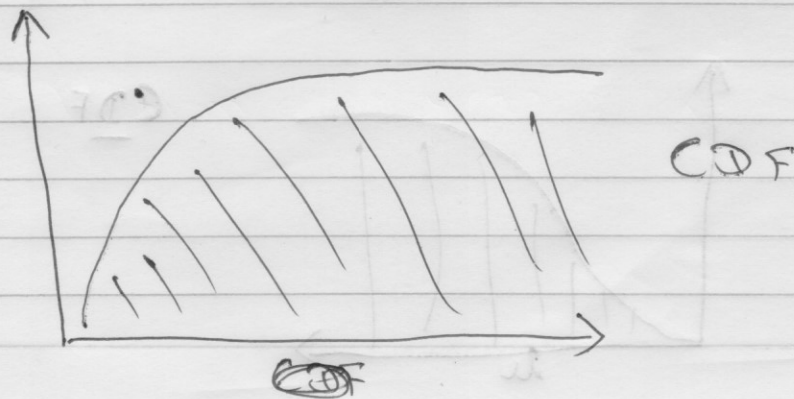
↳ Expected Value $E(Y) = \frac{1}{\lambda}$

↳ Variance $Var(Y) = \frac{1}{\lambda^2}$

↳ Often used in Data Analysis.

↳ Generally, when encountered, people prefer to Transform it in another Distribution.

↳ Generally, we can take natural logarithm of every set of an Exponential Distribution and get a Normal Distribution.



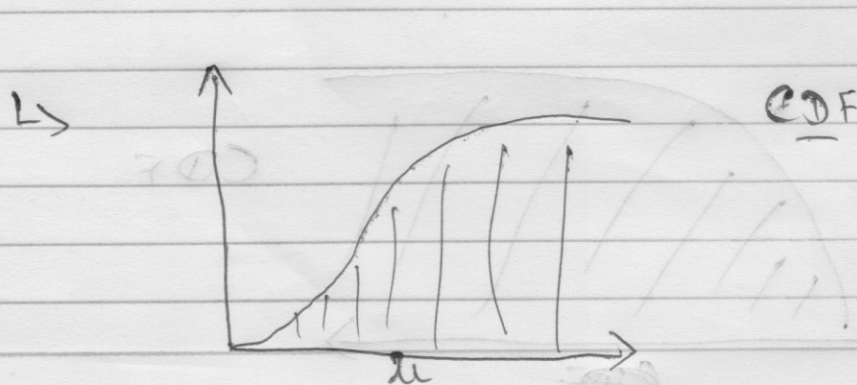
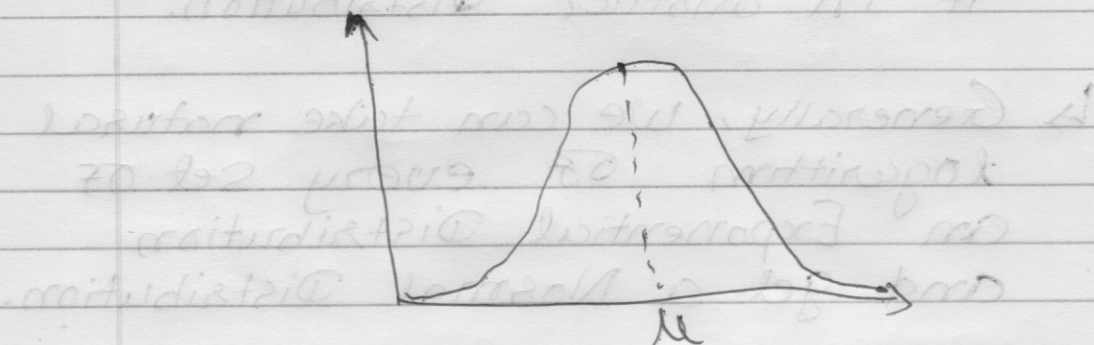
L6) Logistic Distribution:

Denoted by "Logistic"

$Y \sim \text{Logistic}(\mu, S)$; Variable Y
Follows a
Logistic

Distribution with mean or
location ' μ ' and scale parameter
of ' S '.

L> Graph of PDF looks similar
to Normal Distribution.





↳ Expected value $E(Y) = \mu$

↳ Variance $Var(Y) = \frac{5^2 \pi^2}{3}$

↳ CDF picks up when we reach values near the mean.

↳ The Smaller the Scale parameter, the quicker it reaches values close to 1.

↳ Often used in sports to determine how a player's or team's performance can determine the outcome of the match.